Lab 3: Foreign Keys and Simple SQL

Objectives:

- Implement a relation schema on MySQL with `CREATE TABLE` and fill it with `INSERT`.
- Express the various data types in MySQL.
- Demonstrate the referential integrity constraints with `FOREIGN KEY`.
- Show, add, edit and remove rows with `SELECT`, `INSERT`, `UPDATE` and `DELETE`, from a small website.

1. Setup

Write every script in a `.sql` file and run it with the **MySQL client**. A DDL script starts with `DROP DATABASE IF EXISTS company; CREATE DATABASE company; USE company;` so it can be re-run any time; the other scripts start with `USE company;`.

```
mysql -u root -p < problem1_DDL.sql    # build the database and its tables
mysql -u root -p < problem1_DML.sql    # insert your rows
mysql -u root -p company               # type queries by hand and read the result
```

Problem 1: you write the DDL and insert your own rows. Problem 2 (database `library`): you write the DDL, the DML (seed data) and a website; you will use them in next week's discussion.

2. Cheat sheet

```
CREATE TABLE table_name (
  column1 INT          NOT NULL,       -- column type  [NOT NULL]
  column2 VARCHAR(30),
  PRIMARY KEY (column1)
);
ALTER TABLE first_table ADD FOREIGN KEY (foreign_key_field_name)
  REFERENCES second_table(second_table_primary_key);
INSERT INTO table_name (column1, column2) VALUES (1, 'text');
DELETE FROM table_name WHERE column1 = 1;
```

You want	You write
whole number	`INT`
decimal number (salary, price)	`DECIMAL(10,2)`
text	`VARCHAR(50)` ; one letter, such as gender: `CHAR(1)`
date	`DATE`, written in quotes as `'2000-09-01'` (year-month-day)
a column that must have a value	`NOT NULL`
link two tables	`FOREIGN KEY (col) REFERENCES other_table(pk_col)`
range, both ends included	`price BETWEEN 15 AND 20`
combine conditions	`AND`, `OR` (use brackets when you mix them)
change a value	`UPDATE book SET price = 12.50 WHERE book_id = 1`
delete a row	`DELETE FROM book WHERE book_id = 1`

- **Referential integrity:** a foreign key value is either `NULL` or equal to an existing primary key value of the referenced table. MySQL rejects any statement that would break this rule.
- A referenced table must exist before a table can point to it; or add the foreign key later with `ALTER TABLE`.
- Text and dates go in quotes: `'Oxford'`, `'2019-01-01'`. Numbers do not.

PART I · PROBLEM 1 · The COMPANY database

Consider the COMPANY relational database schema:

EMPLOYEE (ssn, fname, lname, bdate, address, gender, salary, Dno)

DEPARTMENT (Dnumber, Dname, mgr_ssn, mgr_start_date)

PROJECT (Pnumber, Pname, Plocation, Dno)

Foreign keys of the schema: `EMPLOYEE.Dno` *and* `PROJECT.Dno` *refer to* `DEPARTMENT.Dnumber`*;* `DEPARTMENT.mgr_ssn` *refers to* `EMPLOYEE.ssn`*.*

Requirements

- Implement the given schema using DDL, in a script run with the MySQL client.
- Insert data in all tables (at least 3 records in each table).
- Answer the following questions:

Q1.1 Insert a row in the DEPARTMENT table that references a row in the EMPLOYEE table. Then try to delete the referenced row. What will happen?

Q1.2 Insert a row in the PROJECT table that references a row in the DEPARTMENT table that doesn't exist yet. What will happen?

Q1.3 If `mgr_ssn` and `Dno` are both NOT NULL, how will you insert the first row in EMPLOYEE or DEPARTMENT?

SQL snippets will help

```
ALTER TABLE first_table ADD FOREIGN KEY (foreign_key_field_name)
  REFERENCES second_table(second_table_primary_key);
```

Remember. For each question write the SQL you used and the output, or the error message if there is one, in `problem1_Answers.txt`.

PART II · PROBLEM 2 · Library: Books and Authors

This library keeps only two tables: its **authors** and their **books**. Implement the schema using DDL statements, with the primary keys (underlined) and the foreign key between the tables, then add it to your **website**:

AUTHOR (author_id, name, country)

BOOK (book_id, title, price, author_id)

Foreign key of the schema: `BOOK.author_id` *refers to* `AUTHOR.author_id`*.* When an author is deleted, all the books of that author are deleted too.

Requirements

- Implement the given schema using DDL, in a script run with the MySQL client.
- Insert **seed data for testing** in both tables (at least 3 authors and 6 books). We will run your DDL and DML in the discussion next week.
- **Continue your website from the last assignment.**

New pages on your website

F2.1 **Show** a table of the books (with the *author_id* of each book) and a table of the authors.

F2.2 **Add** a book and **add** an author.

F2.3 **Edit** a book and **edit** an author.

F2.4 **Remove** a book.

F2.5 **Remove** an author.

> **Note.** You do not have to write relational queries (no JOIN is needed), and you are not required to use advanced filters such as GROUP BY or HAVING. Simple SELECT, INSERT, UPDATE and DELETE statements are enough.

Bonus: upload images for the books (a cover) and for the authors (a photo) on your website.

Lab Deliverables

Deliver the following files, **all of them in the same folder**, in a GitHub repository:

```
lab3-<student-id>/
  problem1_DDL.sql      # DDL scripts for the database creation (Problem 1)
  problem1_DML.sql      # INSERT queries: at least 3 records in each table
  problem1_Answers.txt  # the SQL you used for Q1.1-Q1.3 and the output/error, if any
  problem2_DDL.sql      # DDL scripts for the database creation (Problem 2)
  problem2_DML.sql      # seed data (INSERT): at least 3 authors and 6 books
  website/              # your Lab 2 website, with the new pages (Problem 2)
  README.md             # how to run it, step by step (update your Lab 2 README)
```

> **Submission.** This is the **last lab assignment**. The submission is your **GitHub repository link**, sent through this form: https://forms.gle/HkZicxUB8yUyaA817
> **Deadline: 26/10/2026 at 11:59 PM.** Keep your seed data: it is the data we use when we discuss the assignment next week.

Grading rubric (10 marks)

Item	Marks
Problem 1 DDL: tables, data types, primary and foreign keys	2
Problem 1 DML: at least 3 records in each table	1
Problem 1 answers Q1.1-Q1.3: SQL and output/error	1.5
Problem 2 DDL: tables, data types, primary keys, foreign key	1
Problem 2 DML: seed data (at least 3 authors, 6 books)	0.5
Website: the two tables are shown on your website (F2.1)	1
Website: add and edit books and authors (F2.2, F2.3)	1
Website: remove a book and remove an author (F2.4, F2.5)	1
Bonus: images for books and authors	Bonus
Required file names, README, scripts run without errors, the website runs by following the README	1

Policies:

- You should work individually.
- If 2 or more copies are discovered, all copies will lose all the marks of year work. Hence, it is better to deliver nothing than delivering a copy.
- Late submission is allowed for one week with 80% of the total mark. No late submission is allowed after that.